

Cover Letter to Journal of Magnetism

[SUBMISSION DATE]

Dear Editor,

I submit the research article entitled *Frozen Cross-Dataset Reliability of Healthy-Only PMSM Stator-Fault Detection* for consideration in the *Journal of Magnetism*.

The article examines whether stator-fault alarms fitted and calibrated on healthy currents retain reliability across datasets. It evaluates current asymmetry and harmonic features under changing operating and measurement conditions, using a frozen pipeline and target-only controls.

The exploratory KAIST result was 95.71% fault-block detection with 0/42 healthy alarms. After a pre-reveal internal Git freeze, the primary Log-Euclidean detector detected 96/384 external fault blocks (25.00%), produced 1/32 healthy alarms and attained AUROC 0.6354. Its descriptive two-sided 95% Wilson upper endpoint of 15.74% exceeded the prespecified 12% health gate. The preimplemented target-only MinCovDet comparator reached 70.05% detection with 0/32 healthy alarms at the primary seed, but its gate passed for only three of five seeds. The failed primary result is retained.

All 48 external fault records came from one physical PMSM. The article reports turn-phase confounding, unequal pooled-AUROC load support, time dependence and absence of real-time evidence. The internal Git freeze was not an externally registered preregistration. A secondary parser failure is retained; its repair is post-reveal and non-confirmatory. These findings connect electromagnetic diagnostics with a concrete engineering failure: high same-family detection and late record-any alarms did not demonstrate reliable transfer.

The supplement preserves all comparators, seed variation and bounded diagnostics. The code package includes protocols, fine-grain results and data-acquisition commands. Source-augmentation contrasts describe complete configurations, including different fit-derived feature masks.

I am the sole author. This article has not been published previously and is not under consideration by another journal. The public data were used in accordance with their licenses. [AUTHOR TO CONFIRM BEFORE SUBMISSION: final manuscript approval, responsibility for all content, and any required institutional submission permission.]

This research received no funding. I declare no competing interests. [AUTHOR TO COMPLETE: acknowledgments, if any, and final approval of the AI assistance disclosure. OpenAI Codex assisted code review, evidence checks, English revision and document preparation; author review and final responsibility are required.]

Sincerely,

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